

Total No. of Questions : 8]

SEAT No. :

PC-2844

[Total No. of Pages : 4

[6352]-68

S.E. (Artificial Intelligence and Machine Learning)

DISCRETE MATHEMATICS

(2019 Pattern) (Semester - III) (218541)

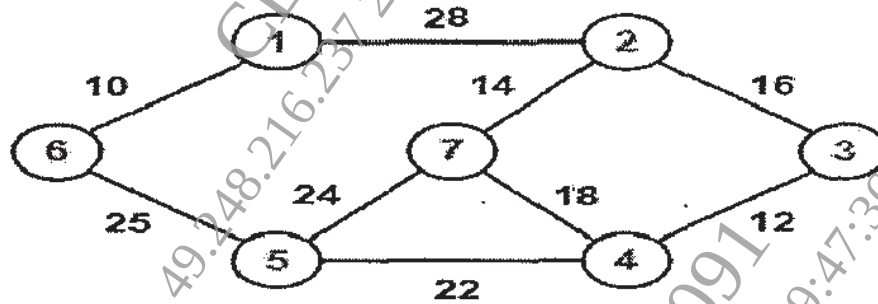
Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1) a) What is a planar graph? Suppose that a connected planar graph has 30 edges. If a planar representation of this graph divides the plane into 20 regions, how many vertices does this graph have? [6]
- b) Build a minimum spanning tree for the following graph using Kruskal's algorithm. [6]

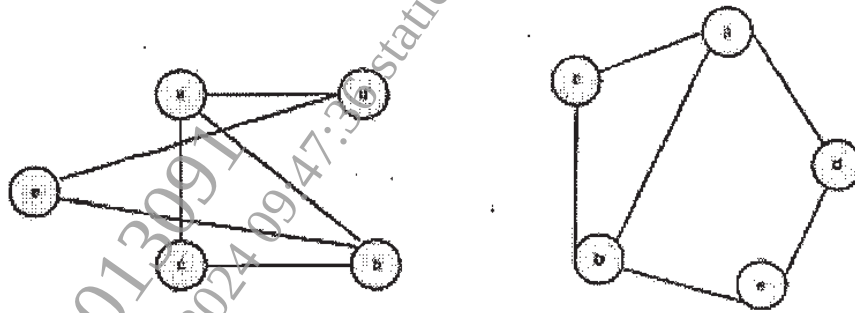


- c) Define Prefix Code. Which of the following codes are prefix codes? Justify your answer. [6]
- i) a: 0, e: 1, t: 01, s: 001
 - ii) a: 101, e: 11, t: 001, s: 011, n: 010

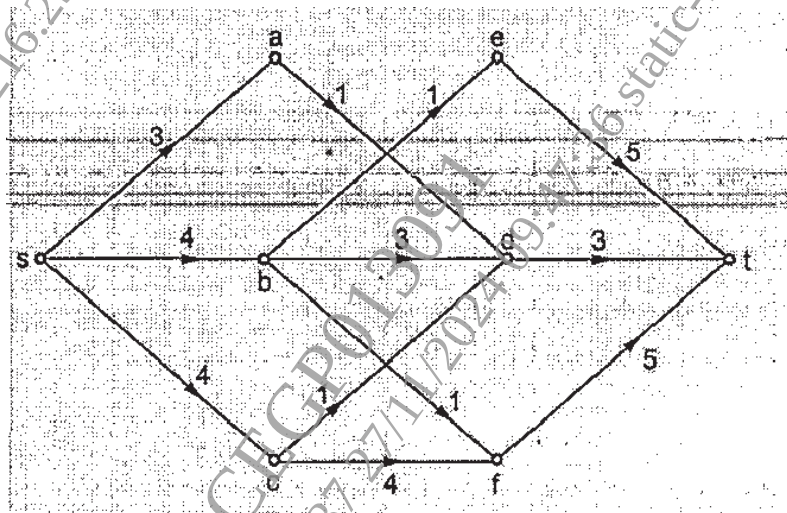
OR

P.T.O.

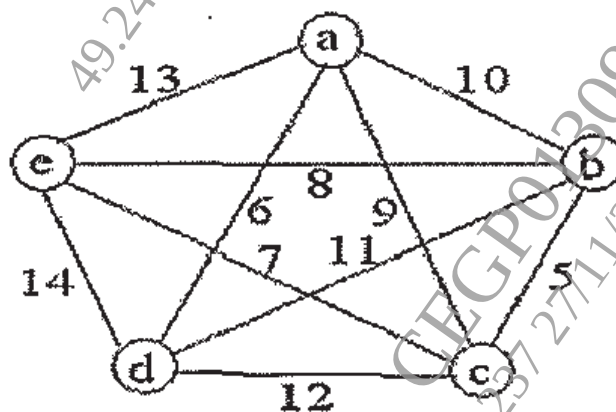
- Q2) a) Determine whether the following graphs are isomorphic to each other. Justify your answer. [6]



- b) Using the labeling procedure, find the maximum flow in the following transport network. [6]



- c) Use the nearest Neighborhood method to solve the Traveling Salesperson problem starting with vertex a. Find the cost of the tour. [6]



Q3) a) Consider these relations on the set of integers : **[6]**

$$R_1 = \{(a, b) \mid a \leq b\},$$

$$R_2 = \{(a, b) \mid a > b\},$$

$$R_3 = \{(a, b) \mid a = b\}$$

$$R_4 = \{(a, b) \mid a = 1 + b\}$$

$$R_5 = \{(a, b) \mid a = b \text{ or } a = -b\},$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}$$

Which of these relations contain each of the pairs (1, 1), (1, 2), (2, 1), (1, -1) and (2, 2)?

b) Solve the following recurrence relation. **[6]**

$$a_n = a_{n-1} + 2.a_{n-2} \text{ where } a_0 = 2, a_1 = 7$$

c) What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D, and F? **[5]**

OR

Q4) a) Explain the injective function with an example? Determine whether each of these functions is an injection from R to R . **[6]**

i) $f(x) = 2x + 1$

ii) $f(x) = x^2 + 1$

iii) $f(x) = x^3$

iv) $f(x) = (x^2 + 1)/(x^2 + 2)$

b) What is POSET? Let A is set of factors of positive integer m and relation is divisibility on A . i.e. $R = \{(x, y) \mid x, y \in A, x \text{ divides } y\}$ For $m = 30$. Draw Hasse Diagram. **[6]**

c) Find the transitive closure by using Warshall's algorithm for the given relation as: **[5]**

$$R = \{(2, 1), (2, 3), (3, 1), (3, 4), (4, 1), (4, 3)\}$$

Q5) a) Using Binary expansion method solve the following (Show step-wise answer) $7^{61} \bmod 15$. **[6]**

b) Which of the following is true? Justify your answer **[6]**

i) $556 \equiv 1296 \pmod{25}$

ii) $1655 \equiv 935 \pmod{23}$

iii) $448 \equiv 788 \pmod{10}$

c) Using Chinese Remainder Theorem, find the value of p using following data **[6]**

$$P \equiv 2 \pmod{11}$$

$$P \equiv 3 \pmod{7}$$

OR

Q6) a) What is a Mersenne prime number? Which of the following is the Mersenne Prime number? 71, 31, 255, 63, 7 Justify your answer. [6]

b) Find the Euler's totient function of the following numbers : [6]

i) 47

ii) 45

iii) 65

c) Find the multiplicative inverse of 1234 mod 4321 using Extended Euclidean Algorithm. [6]

Q7) a) Consider the set $A = \{1, 3, 5, 7, 9, \dots\}$ i.e. a set of odd positive integers. Determine whether A is closed under: [6]

i) $a*b = a + 3b$

ii) $a*b = (a-b) / 2$.

iii) $a*b = \max(a, b)$

iv) $a*b = \text{power}(a, b)$

v) $a*b = 2a-b$

vi) $a*b = \text{GCD}(a, b)$

b) Show that $(\mathbb{Z}_6, +)$ is an Abelian Group. [6]

c) Prove that Hamming Distance $d(x,y) = d(y,x)$. [5]

OR

Q8) a) Show that $G = \{1, w, w^2\}$ is an abelian group under multiplication where w is the cube root of unity. [6]

b) Explain Integral Domain with an example. [6]

c) Consider the (2,6) encoding function e. $e(00) = 110000$, [5]

$e(10) = 101000$

$e(01) = 011110, e(11) = 111001$

Find the minimum distance of e.
